Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **AWS-based integrations**, event-driven services, and production APIs using **JavaScript, TypeScript, Node.js, Python**, automated testing, CloudWatch monitoring, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Architected a **multi-tenant integration platform** with normalized schemas, row-level isolation, and indexed SQL queries, reducing dashboard latency from 1.2 seconds to 350 milliseconds while supporting 500+ users.
- Built API and data integrations across FastAPI, SQL Server, PostgreSQL, and cloud storage, processing 5K+ monthly jobs with **payload validation, error handling, monitoring, and CI/CD** while maintaining 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Delivered **React and Flask integration features** for uploads, grading history, and feedback, supporting 150+ instructors and 10K+ daily requests through maintainable REST APIs and reliable production workflows.
- Implemented multithreaded processing and automatic fallbacks for 1K+ files per batch, reducing turnaround 60% while validating changes through **functional testing, debugging, code reviews, and CI/CD** releases safely.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built **AWS Glue and S3 integration pipelines** for JSON, CSV, and Parquet data, improving processing efficiency 68% while using object-oriented patterns to reduce API latency 30% and technical debt 20%.
- Automated 20+ **unit, integration, and regression tests** using PyTest, Mockito, and Jenkins across Windows, Linux, and macOS, collaborating through Agile reviews and Git workflows to improve release confidence.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **Node.js and React integration services** with REST APIs, WebSockets, authentication, billing, permissions, and notifications, synchronizing 2M+ records across distributed locations through reliable asynchronous event-driven production workflows.
- Owned eight services through implementation and deployment, adding **automated testing, logging, monitoring, Docker, and Jenkins CI/CD** that cut release time 50% while sustaining 95%+ reliability in production.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built **React and TypeScript integration workflows** for booking, pricing, coupons, and payment-status updates, serving 50K+ records through Flask APIs with sub-200-millisecond responses and JWT-secured access across customer workflows.
- Designed 12+ **API and webhook integrations** for payments, synchronization, and status events, documenting workflows and incorporating client feedback before implementation to reduce design errors 25% across delivery cycles.

Skills

Languages: JavaScript, TypeScript, Node.js, Python, Java, SQL, HTML5, CSS3, JSON, XML, Bash, Shell
AWS & Integration: AWS Lambda, SQS, SNS, S3, API Gateway, CloudWatch, AWS Glue, REST APIs, Webhooks, Event-Driven Systems, Messaging, Data Flows
Applications & Data: React.js, Flask, FastAPI, Microservices, PostgreSQL, SQL Server, MySQL, MongoDB, Firebase, Schema Design, Query Optimization
Testing & Delivery: Functional Testing, Unit Testing, Integration Testing, Regression Testing, Automated Testing, PyTest, Mockito, Jest, Git, JIRA, Jenkins, GitHub Actions, CI/CD, Docker
Engineering: Payload Validation, Error Handling, Logging, Monitoring, Troubleshooting, Root-Cause Analysis, Production Support, Code Reviews, Technical Documentation, Agile/Scrum, Release Management

Projects

EventBridge Hub | *TypeScript, Node.js, AWS Lambda, SQS, SNS, API Gateway, CloudWatch*

- Designed an **AWS event-driven integration hub** using TypeScript, Lambda, SQS, SNS, and API Gateway, with schema validation, dead-letter handling, idempotent processing, structured logging, and CloudWatch monitoring for resilience.

IntegrationCheck Harness | *Node.js, TypeScript, Jest, REST APIs, S3, GitHub Actions*

- Built an **automated integration-testing harness** for APIs and event flows, validating payloads, retries, error paths, and expected outcomes while publishing regression results and diagnostic artifacts through CI workflows.

Linguistix | *Python, AWS ECS, Fargate, CodePipeline, Docker, REST APIs*

- Developed and deployed a containerized **AWS data-processing API** with validation checks, Docker, ECS Fargate, and CodePipeline automation, achieving 87% accuracy while supporting repeatable CI/CD releases and reliable operations.